

$$\text{例 } J_\nu(x) = \sum_{m=0}^{\infty} (-1)^m \frac{x^{\nu+2m}}{2^{\nu+2m} m! \Gamma(\nu+m+1)}$$

$$\textcircled{2} C = -\nu.$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} (-1)^m \frac{x^{-\nu+2m}}{2^{-\nu+2m} m! \Gamma(-\nu+m+1)}$$

$$y = C_1 J_\nu(x) + C_2 J_{-\nu}(x)$$

$\nu = 2 \Rightarrow J_2$ 求线性无关的另一个解

$$y_2(x) = \frac{J_2 \cos 2x - J_2(x)}{\sin 2x}$$

$$x^2 y'' + xy' + (x^2 - 2^2)y = 0.$$

第三章

$$\S 1 \quad \frac{dx_i}{dt} = f_i(t, x_1, \dots, x_n)$$

$$x_1(t) \dots x_n(t)$$

$$\frac{dx_i(t)}{dt} = f_i(t, x_1(t), \dots, x_n(t))$$

$$\frac{d^n x}{dt^n} = f(t, x, \frac{dx}{dt}, \dots, \frac{d^{n-1}x}{dt^{n-1}})$$

$$\text{令 } x = x_i \quad \frac{dx_i}{dt} = x_2 \quad \frac{dx_{n-1}}{dt} = x_n$$

$$\begin{cases} \frac{dx_1}{dt} = x_2 \\ \frac{dx_2}{dt} = x_3 \\ \vdots \\ \frac{dx_{n-1}}{dt} = x_n \end{cases}$$

$$\frac{dx_n}{dt} = f(t, x_1, \dots, x_n)$$

$$x_1(t_0) = x_{10}$$

$$x_n(t_0) = x_{n0}$$

$$x_i = \varphi_i(t, c_1, \dots, c_n)$$

$$\dot{\Phi}_i(t, x_1, \dots, x_n; c_1, \dots, c_n) = 0$$

$$x_i = \varphi_i(t, c_1, \dots, c_n) \text{ 通积分}$$

$$\frac{dx_i}{dt} = \sum_{j=1}^n a_{ij}(t)x_j + f_i(t)$$

$$\frac{d^n x}{dt^n} + p_1(t) \frac{d^{n-1}x}{dt^{n-1}} + \dots + p_n(t)x = f(t) \quad (1)$$

$$\text{令 } x = x_1 \quad \frac{dx_1}{dt} = x_2 \quad \frac{dx_{n-1}}{dt} = x_n$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \frac{dx}{dt} = \begin{bmatrix} \frac{dx_1}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{bmatrix} \quad f(t) = \begin{bmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{bmatrix}$$

$$A(t) = \begin{bmatrix} a_{11}(t) & \dots & a_{1n}(t) \\ \vdots & & \vdots \\ a_{n1}(t) & \dots & a_{nn}(t) \end{bmatrix}$$

$$(1) \Rightarrow \frac{dx}{dt} = A(x) \times f(t)$$

$$f(t) \neq 0 \quad \text{非齐次}$$

$$\frac{dx}{dt} = A(t) \times \text{齐次}$$

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

§2.

$$\begin{cases} \frac{dx}{dt} = A(t)x + f(t) \\ x(t_0) = x_0 \end{cases}$$

$$\begin{cases} \frac{dx}{dt} = A(t)x \\ x(t_0) = 0. \end{cases}$$

$$x(t) \equiv 0.$$

0 解.

$$x = c_1 x_1(t) + \dots + c_m x_m(t)$$

$$\frac{d}{dt} \sum c_i x_i(t) = A(t) \sum c_i x_i(t)$$

$$= \sum c_i \frac{dx_i(t)}{dt} = \sum c_i A(t) x_i(t)$$

$$= \sum c_i \left[\frac{dx_i(t)}{dt} - A(t) x_i(t) \right] \\ \begin{matrix} 0 \\ 0 \end{matrix}$$

定义 $a_1 x_1(t) + \dots + a_m x_m(t) \equiv 0$. 线性相关.

若仅 $a_1 = \dots = a_m = 0$ 成立, 线性无关.

$\gamma_1 e^{\lambda_1 t} \dots \gamma_m e^{\lambda_m t}$. 线性无关.

$$a_1 \gamma_1 + a_2 \gamma_2 e^{(\lambda_2 - \lambda_1)t} + \dots + a_n \gamma_n e^{(\lambda_n - \lambda_1)t} \equiv 0$$

$$\lambda_i - \lambda_1 \neq 0.$$

$$\beta_m \gamma_m e^{\lambda_m t} \equiv 0.$$

$$x(t) = (x_1(t) \dots x_n(t))$$

$\det x(t)$. 朗斯基行列式

$$W(t) = \det X(t) = \begin{vmatrix} x_{11}(t) & \cdots & x_{1n}(t) \\ \vdots & & \vdots \\ x_{n1}(t) & \cdots & x_{nn}(t) \end{vmatrix}$$

$x_1(t) \cdots x_n(t)$. 齐次线性方程组 n 个解.

$x_1(t) \cdots x_n(t)$ 相关 $\Leftrightarrow W(t) \equiv 0$.

$$X(t) = \sum c_i x_i(t),$$

$x_i(t)$ 为线性无关解. 基本解组

基本矩阵

$$x(t) = X(t)C$$

$$W(t_0) = \begin{vmatrix} 1 & & 0 \\ & 1 & \\ 0 & & 1 \end{vmatrix} = 1 \neq 0.$$

$$x = X(t)C + x^*(t)$$

§3 常系数

$$\frac{dx}{dt} = Ax + f(t)$$

$$\frac{dx}{dt} = Ax$$

$$x = ve^{\lambda t}$$

$$\lambda e^{\lambda t} = A v e^{\lambda t}$$

$$E = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$(A - \lambda E)v = 0.$$

$$\det(A - \lambda E) = 0.$$

$$D(\lambda) = \begin{vmatrix} a_{11} - \lambda & \dots & a_{1n} \\ \vdots & a_{22} - \lambda & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix} = 0.$$

$$x(t) = v_k e^{\lambda_k t}.$$

$\lambda = \lambda_k$, 特征根.

v_k 为 A 属于 λ_k 特征向量.

$$x(t) = v_k e^{\lambda_k t}$$

(-) 单根

$$v_1 e^{\lambda_1 t}, \dots, v_n e^{\lambda_n t}$$

$$x = \sum c_i v_i e^{\lambda_i t}$$

eg.
$$\begin{cases} \frac{dx_1}{dt} = -3x_1 + 4x_2 - 2x_3, \\ \frac{dx_2}{dt} = x_1 + x_3, \\ \frac{dx_3}{dt} = 6x_1 - 6x_2 + 5x_3. \end{cases}$$

$$\begin{vmatrix} -3-\lambda & 4 & -2 \\ 1 & -\lambda & 1 \\ 6 & -6 & 5-\lambda \end{vmatrix}$$

$$= -(\lambda-2)(\lambda-1)(\lambda+1) = 0.$$

$\lambda_1=2/\lambda_2=1/\lambda_3=-1$ 单根.

$$\text{设 } \lambda_1=2 \quad v_1 = \begin{bmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{bmatrix}$$

$$\begin{bmatrix} -3-\lambda_1 & 4 & -2 \\ 1 & -\lambda_1 & 1 \\ 6 & -6 & 5-\lambda_1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{bmatrix} = 0.$$

$$\begin{bmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} e^{2t}$$

$$\lambda_2: \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \lambda_3: \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t + c_3 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-t}$$

若为复数根

$$\lambda_1 = \alpha + i\beta \quad \lambda_2 = \alpha - i\beta$$

$$v_1 = p + iq \quad v_2 = \bar{v}_1 = p - iq$$

$$v_1(t) = (p + iq)e^{(\alpha + i\beta)t}$$

$$= e^{\alpha t} (p \cos \beta t - q \sin \beta t) + i e^{\alpha t} (p \sin \beta t + q \cos \beta t)$$

$$x_2(t) = (p - iq) e^{(\alpha - i\beta)t}$$

$$= e^{\alpha t} (p \cos \beta t - q \sin \beta t) - i e^{\alpha t} (p \sin \beta t + q \cos \beta t)$$

↙ ↘
两个实值解.

eg. $\frac{d}{dt} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$

$$D(\lambda) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 3-\lambda & -1 \\ -1 & 2 & 3-\lambda \end{vmatrix}$$

$$= -(\lambda-2)(\lambda^2-6\lambda+10)$$

$$\lambda_1 = 2 \quad \lambda_{2,3} = 3 \pm i$$

$$\begin{vmatrix} \cos & -\sin \\ \sin & \cos \end{vmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\alpha = 3 \quad \beta = 1$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

$$\begin{bmatrix} 1 \\ 1+i \\ 2-i \end{bmatrix} e^{(3+i)t}$$

$$= \begin{bmatrix} \cos t \\ \cos t - \sin t \\ 2 \cos t + \sin t \end{bmatrix} e^{3t} + i \begin{bmatrix} \sin t \\ \cos t + \sin t \\ 2 \sin t - \cos t \end{bmatrix} e^{3t}$$

$$C_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + C_2 \begin{bmatrix} \cos t \\ \cos t - \sin t \\ 2\cos t + \sin t \end{bmatrix} e^{3t} + C_3 \begin{bmatrix} \sin t \\ \cos t + \sin t \\ 2\sin t - \cos t \end{bmatrix} e^{3t}$$

(二) 有重根

$$x(t) = (v_0 + \frac{t}{1!} v_1 + \frac{t^2}{2!} v_2 + \dots + \frac{t^{k-1}}{(k-1)!} v_{k-1}) e^{\lambda_0 t}$$

$$(A - \lambda_0 E)^k v_0 = 0$$

$$\frac{dx}{dt} = Ax$$

$$\lambda_1, \dots, \lambda_s \quad n_1 + \dots + n_s = n$$

$$p_i(t) e^{\lambda_i t}$$

$p_i(t)$ 向量函数.

$$x_i = \sum_{j=1}^s \sum_{l=1}^{n_j} c_{ij} p_j(t) e^{\lambda_j t}$$

例 3 解方程组

eg. $\frac{dx}{dt} = Ax, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 4 \\ 1 & 0 & -4 \end{bmatrix}.$

$$\det(A - \lambda E)$$

$$= \begin{vmatrix} -1-\lambda & 1 & 0 \\ 0 & -1-\lambda & 4 \\ 1 & 0 & -4-\lambda \end{vmatrix}$$

$$= -\lambda(\lambda+3)^2 = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = -3 \text{ 重根}$$

$$\lambda_1 = 0$$

$$V_0 = \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix}$$

$$(A+3E)^2 V_0 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 4 \\ 1 & 0 & -1 \end{bmatrix}^2 \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix} = 0.$$

$$V_0^{(1)} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad V_0^{(2)} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$V_1^{(1)} = (A+3E) V_0^{(1)} \\ = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 4 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_2 = \left[\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} -2 \\ -4 \\ 2 \end{bmatrix} \right] e^{-3t} (V_0^{(1)} + t V_1^{(1)}) e^{-3t}$$

$$V_1^{(2)} = (A+3E) V_0^{(2)} \\ = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 4 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_3 = \left[\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} \right] e^{-3t}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = c_1 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_1 + c_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_2 + c_3 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_3 \\ = c_1 \begin{pmatrix} 4 \\ 4 \\ 1 \end{pmatrix} + c_2 \left[\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} -2 \\ -4 \\ 2 \end{pmatrix} \right] e^{-3t} \\ + c_3 \left[\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \right] e^{-3t}.$$

eg.
$$\begin{cases} \frac{dx}{dt} = -x + y + z \\ \frac{dy}{dt} = x - y + z, \\ \frac{dz}{dt} = x + y - z. \end{cases}$$

$$\begin{vmatrix} -1-\lambda & 1 & 1 \\ 1 & -1-\lambda & 1 \\ 1 & 1 & -1-\lambda \end{vmatrix}$$

$$-(\lambda-1)(\lambda+2)^2$$

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t \quad \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

二. 非齐次

变通任意常数法

$$\frac{dx}{dt} = A(t)x + f(t)$$

$$\frac{dx}{dt} = A(t)x$$

$$x = \chi(t) c.$$

$$x = \chi(t) c(t).$$

$$\frac{d(x)}{dt} = \frac{d\chi(t)}{dt} c(t) + \chi(t) \frac{d(c(t))}{dt}.$$

$$= A(t) \chi(t) c + \chi(t) \frac{d(c(t))}{dt}.$$

$$\chi(t) c'(t) = f(t)$$

$$c'(t) = X^{-1}(t) = f(t)$$

$$c(t) = \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

$$c(t_0) = 0$$

$$x^* = X(t) = \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

$$x(t) = X(t)\vec{c} + X(t) \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

其中 $\vec{c}$ 为 n 维任意常向量.

$$t_0 \in (a, b)$$

$$\therefore \text{pre. } x(t_0) = x_0. \text{ 代入}$$

$$\Rightarrow \vec{c} = X^{-1}(t_0)x_0.$$

$$\begin{cases} \frac{dx}{dt} = A(t)x + f(t), \\ x(t_0) = x_0, t_0 \in (a, b) \end{cases}$$

$$\text{sol. } x(t) = X(t)X^{-1}(t_0)x_0 + X(t) \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

$$t_0 \in (a, b)$$

例 6 试解初值问题

eg. $\frac{dx}{dt} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ e^t \cos 2t \end{pmatrix},$

$$x(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 2 & 1-\lambda & -2 \\ 3 & 2 & 1-\lambda \end{vmatrix} = (1-\lambda)(\lambda^2 + 2\lambda + 5) = 0$$

$$\lambda_1 = 1, \lambda_2 = 1+2i, \lambda_3 = 1-2i.$$

$$\Downarrow$$

$$\begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix} \quad x_1 = \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix} e^t$$

$$\lambda_2: v_2 = \begin{bmatrix} 0 \\ 1 \\ -i \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 1 \\ -i \end{bmatrix} e^{(1+2i)t}.$$

$$= e^t \begin{bmatrix} 0 \\ 1 \\ -i \end{bmatrix} (\cos 2t + i \sin 2t)$$

$$= e^t \left[\begin{bmatrix} 0 \\ \cos 2t \\ \sin 2t \end{bmatrix} + i \begin{bmatrix} 0 \\ \sin 2t \\ -\cos 2t \end{bmatrix} \right]$$

$$X(t) = e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix}$$

$$e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix} c'(t) = \begin{bmatrix} 0 \\ 0 \\ e^t \cos 2t \end{bmatrix}$$

$$C'(t) = \begin{bmatrix} 0 \\ \sin 2t \cos 2t \\ -\cos^2 2t \end{bmatrix}$$

$$C = C' \int_0^t \begin{bmatrix} 0 \\ \sin 2\tau \cos 2\tau \\ -\cos^2 2\tau \end{bmatrix} d\tau$$

$$= Ct \begin{bmatrix} 0 \\ \frac{1}{8}(1 - \cos 4t) \\ -\frac{1}{2} - \frac{1}{8}\sin 4t \end{bmatrix}$$

$$\Rightarrow x(t) = e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix} \left[c + \begin{bmatrix} 0 \\ \frac{1}{8}(1 - \cos 4t) \\ -\frac{t}{2} - \frac{1}{8}\sin 4t \end{bmatrix} \right].$$

再由 $x(0) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, 得

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix} c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

所以

$$c = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

于是所给初值问题的解是

$$\begin{aligned} x(t) &= e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{8}(1 - \cos 4t) \\ -1 - \frac{t}{2} - \frac{1}{8}\sin 4t \end{bmatrix} \\ &= e^t \begin{bmatrix} 0 \\ \cos 2t - \left(1 + \frac{t}{2}\right) \sin 2t \\ \left(1 + \frac{t}{2}\right) \cos 2t + \frac{5}{4} \sin 2t \end{bmatrix}. \end{aligned}$$

eg.

例 7 求方程组

$$\begin{cases} \frac{dx}{dt} = x + 2y + e^t, \\ \frac{dy}{dt} = 4x + 3y \end{cases} \rightarrow y = \frac{1}{2} \left(\frac{dx}{dt} - x - e^t \right)$$

$$\frac{1}{2} \frac{d}{dt} \left(\frac{dx}{dt} - x - e^t \right) = 4x + \frac{3}{2} \left(\frac{dx}{dt} - x - e^t \right)$$

$$\frac{d^2x}{dt^2} - 4 \frac{dx}{dt} - 5x = -2e^t$$

$$x = c_1 e^{5t} + c_2 e^{-t} + \frac{1}{4} e^t$$

$$\Rightarrow y = 2c_1 e^{5t} - c_2 e^{-t} - \frac{1}{2} e^t$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{5t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} + \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{2} \end{pmatrix} e^t.$$

例 8 求方程组

eg.

$$\begin{cases} 2 \frac{dx}{dt} + \frac{dy}{dt} + y - t = 0, \\ \frac{dx}{dt} + \frac{dy}{dt} - x - y - 2t = 0 \end{cases}$$

$$\frac{dx}{dt} + x + 2y + t = 0.$$

$$\Rightarrow y = -\frac{1}{2} \left(\frac{dx}{dt} + x + t \right)$$

$$\text{代入} \Rightarrow \frac{d^2x}{dt^2} - 2 \frac{dx}{dt} + x = -3t - 1,$$

